RoboPIN

Grounded Embodied Reasoning via Pinned Chain-of-Thought

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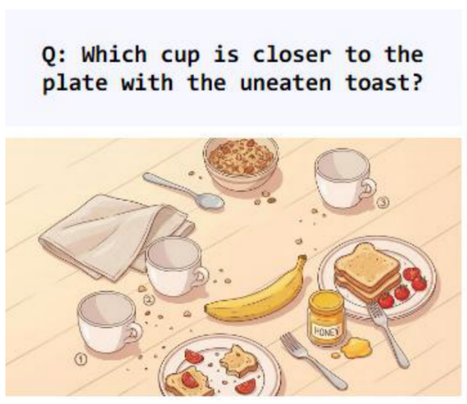
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Motivation



Q: Which cup is closer to the plate with the uneaten toast?

RoboBrain2 (Text CoT)

...the cup positioned closer to the plate is indeed the one further back on the table... Therefore, the cup ... is the one positioned further back. So the answer is ①.

✗ Ambiguous reference

VoCoT (Coord CoT)

Find the cup <coord> 0.166,0.125,0.833,0.999</coord>. Check the plate in the back of it, got the plate <coord> 0.463,... So the answer is cup ①.

✗ Disconnected answer

RoboPIN (PinCoT)



First, locate the plate with the untoasted bread... Next, find the three cups obj1 obj2 obj3

We can judge by spatial position and visual distance: obj3 is adjacent to the plate and is the closest one; obj2 is in the middle, farther away... So the answer is cup ③

✓ Consistent with Pinned Reasoning Anchor

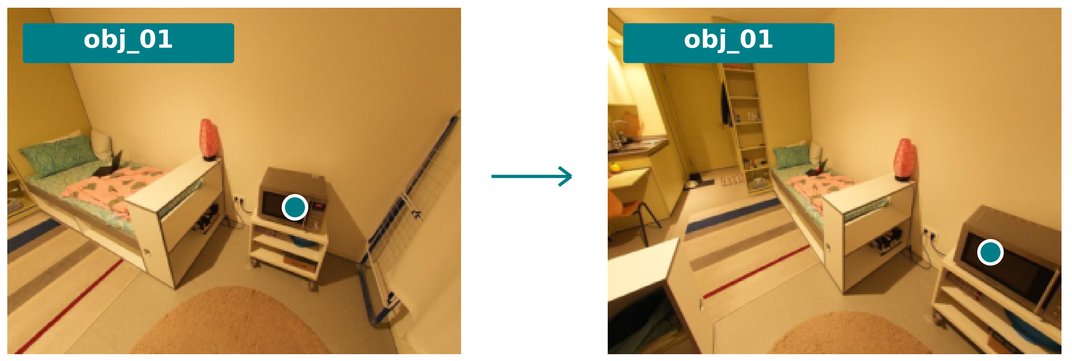
1 Reference drift

Textual references can switch between similar objects during multi-step reasoning.

2 Reasoning–answer misalignment

A correct intermediate analysis can still end on the wrong object, especially across views.

Key Insight: Persistent Identity



Same entity. Same ID. Different views.

Reasoning anchor = name + ID + view + point.

Introduce the anchor; reuse its identity throughout the reasoning chain.

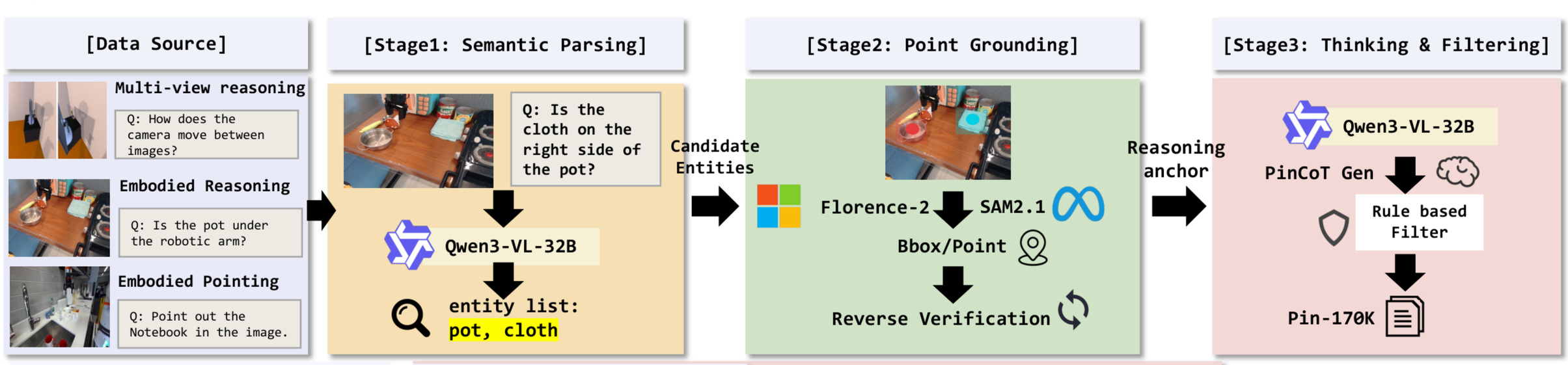
Our insight

Persistent IDs connect visual grounding, intermediate steps and the final answer.

QR PLACEHOLDER

PAPER / CODE

Our Method: RoboPIN / PinCoT



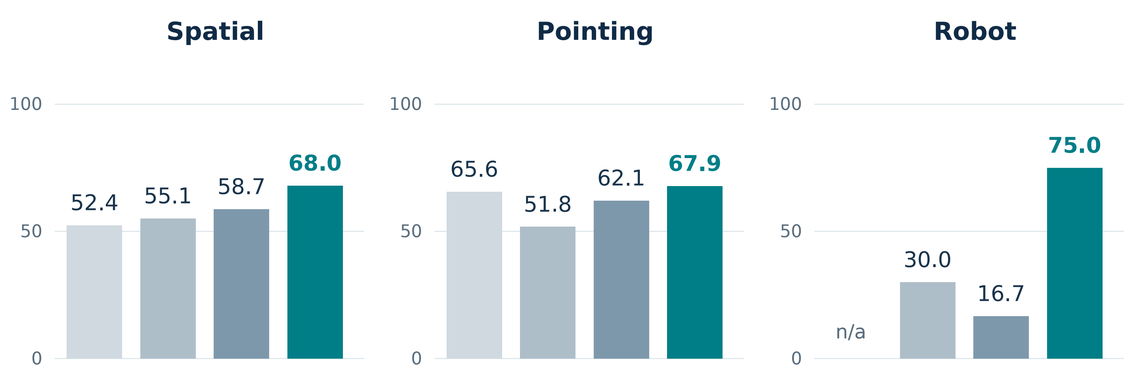
Three-stage post-training

SFT — Embodied knowledge → **CoT-SFT** — Structured reasoning → **RFT / GRPO** — Process alignment

PinCoT binds visual evidence to persistent identities. PIN-170K supplies structured traces; process rewards supervise grounding and identity consistency.

Experimental Results

Embodied-R1 (3B) · RoboBrain2.0 (7B) · Mimo-Embodied (7B) · **RoboPIN (4B)**

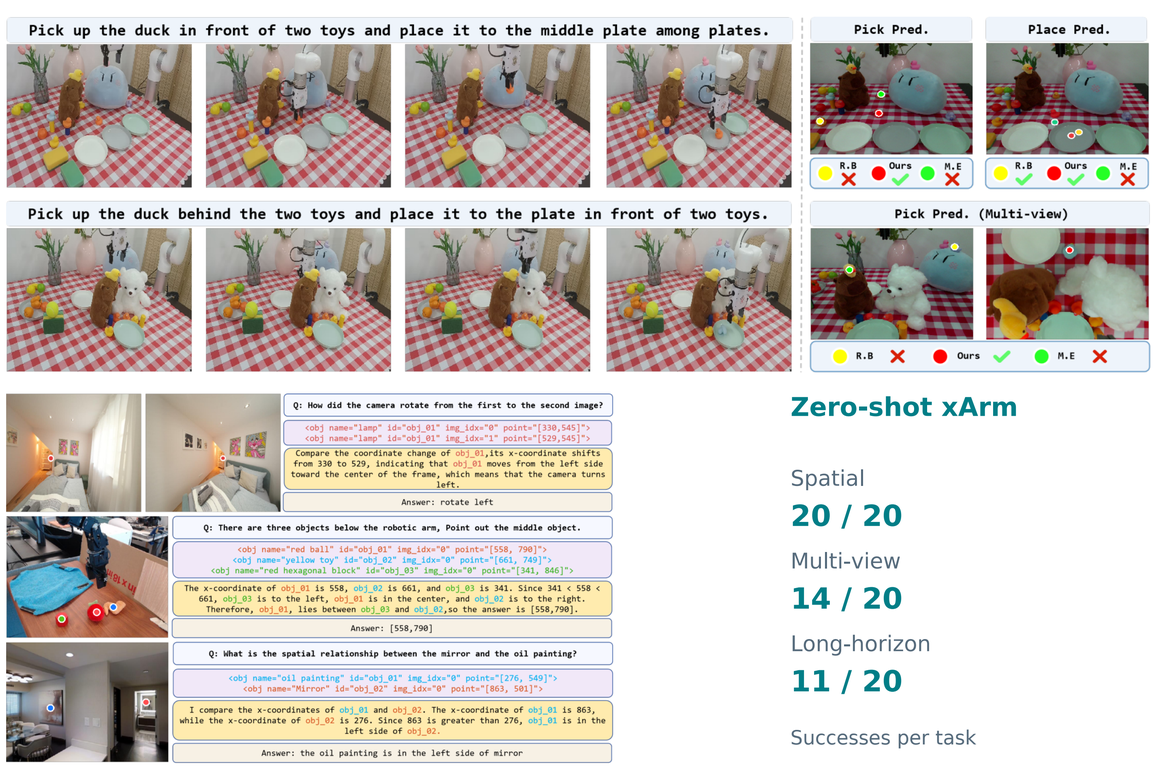


Scores (%) · Spatial: 13-column mean; pointing: 5-benchmark mean; robot: success over 60 trials.

18.8% → 99.9% cross-step ID coverage

Without RFT → RoboPIN · EmbSpatial

Qualitative Results



Zero-shot xArm

Spatial
20 / 20

Multi-view
14 / 20

Long-horizon
11 / 20

Successes per task

Persistent visual anchors make embodied reasoning traceable.